

# **MMB**

*Morsbach Maschinenbau GmbH.*

## **MMB 4" SUBMERSIBLE PUMP 4S-SERIES CATALOGUE**



# STAINLESS STEEL SUBMERSIBLE PUMPS

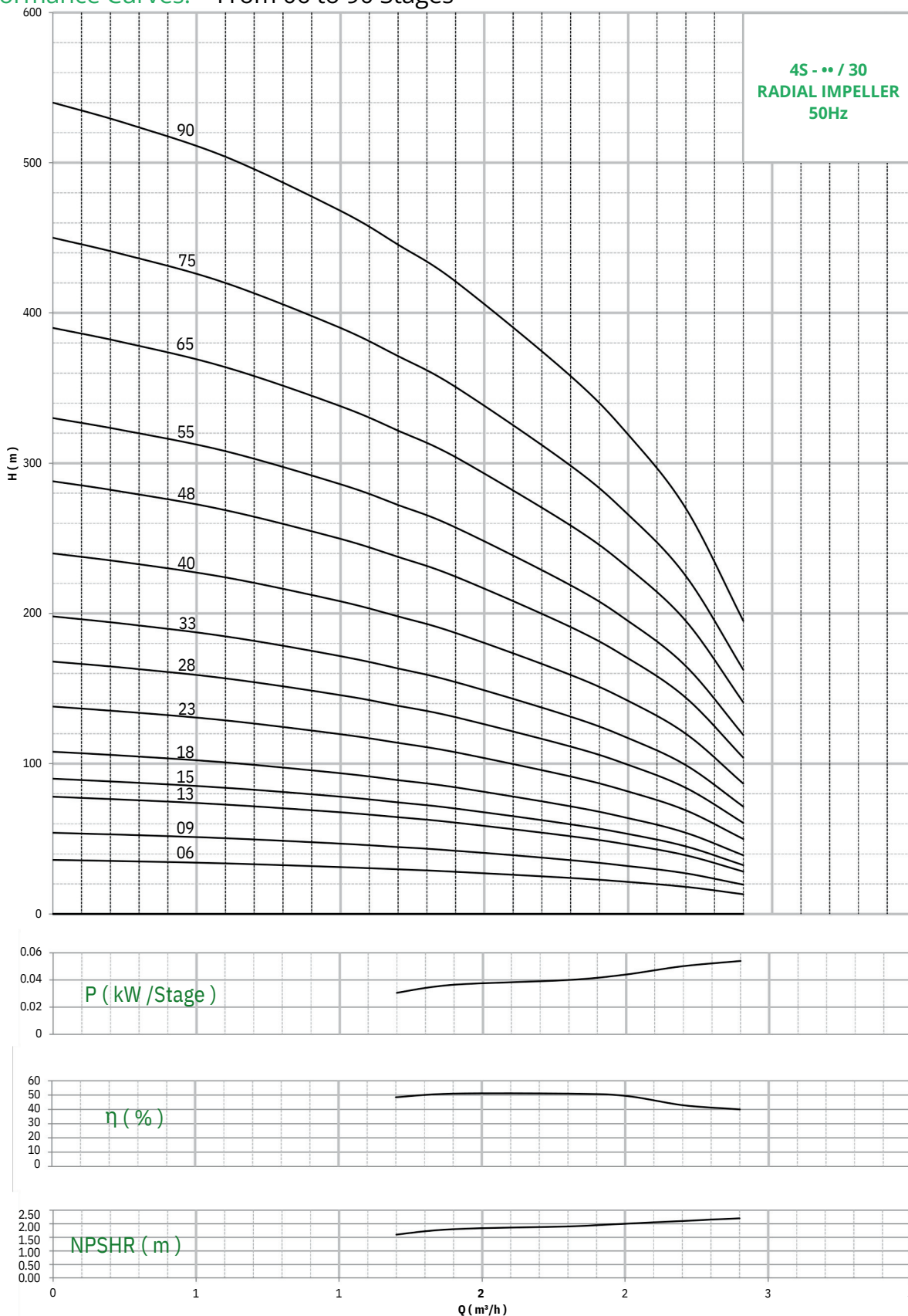
Nominal Diameter: **4"**

Nominal Flow: **2m<sup>3</sup>/h**

Outlet Size: **1.25"**

Flow Range: **1.4-2.2m<sup>3</sup>/h**

Performance Curves: From 06 to 90 Stages



Refer General Information for performance curve conditions and for other details. Curves tolerance according to ISO : 9906, Annex - A

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# STAINLESS STEEL SUBMERSIBLE PUMPS

Nominal Diameter: **4"**

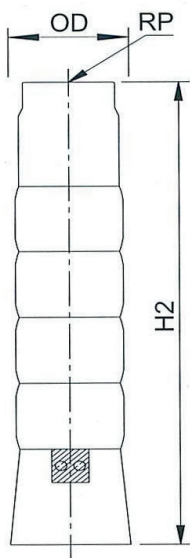
Nominal Flow: **2m<sup>3</sup>/h**

Outlet Size: **1.25"**

Flow Range: **1.4-2.2m<sup>3</sup>/h**

Performance Curves : From 6 to 90 Stages

DIMENSIONS & WEIGHT



| PUMP MODEL | No. of Stages | REQUIRED MOTOR POWER |      | APPROX. DIMENSIONS ( mm ) |    |      | APPROX NETT. WEIGHT ( kg ) |
|------------|---------------|----------------------|------|---------------------------|----|------|----------------------------|
|            |               | Kw                   | HP   | #OD                       | RP | H2   |                            |
| 4S-06/30   | 6             | 0.37                 | 0.5  | 97                        | 32 | 292  | 2.8                        |
| 4S-09/30   | 9             | 0.37                 | 0.5  | 97                        | 32 | 357  | 3.3                        |
| 4S-13/30   | 13            | 0.55                 | 0.75 | 97                        | 32 | 445  | 4.5                        |
| 4S-15/30   | 15            | 0.55                 | 0.75 | 97                        | 32 | 489  | 5.1                        |
| 4S-18/30   | 18            | 0.75                 | 0.75 | 97                        | 32 | 554  | 5.7                        |
| 4S-23/30   | 23            | 1.1                  | 1    | 97                        | 32 | 665  | 7.1                        |
| 4S-28/30   | 28            | 1.5                  | 1.5  | 97                        | 32 | 775  | 8.2                        |
| 4S-33/30   | 33            | 1.5                  | 2    | 97                        | 32 | 1022 | 12.0                       |
| 4S-40/30   | 40            | 2.2                  | 2    | 97                        | 32 | 1214 | 14.3                       |
| 4S-48/30   | 48            | 2.2                  | 3    | 97                        | 32 | 1382 | 16.9                       |
| 4S-55/30   | 55            | 3                    | 4    | 97                        | 32 | 1550 | 19.7                       |
| 4S-65/30   | 65            | 3                    | 4    | 97                        | 32 | 1838 | 22.1                       |
| 4S-75/30   | 75            | 4                    | 5.5  | 97                        | 32 | 2078 | 25.8                       |
| 4S-90/30   | 90            | 4                    | 5.5  | 97                        | 32 | 2438 | 30.0                       |

| PUMP MODEL | No. of Stages | MOTOR ( Kw ) | lps                                                | 0    | 0.07 | 0.17 | 0.28 | 0.33 | 0.39 | 0.50 | 0.56 | 0.61 | 0.67 |
|------------|---------------|--------------|----------------------------------------------------|------|------|------|------|------|------|------|------|------|------|
|            |               |              | m <sup>3</sup> /h                                  | 0.00 | 0.25 | 0.60 | 1.00 | 1.20 | 1.40 | 1.80 | 2.00 | 2.20 | 2.40 |
| 4S-06/30   | 6             | 0.37         | TOTAL MANOMETRIC HEAD IN METERS - ( COLUMN WATER ) | 36   | 35   | 34   | 31   | 30   | 28   | 24   | 21   | 18   | 13   |
| 4S-09/30   | 9             | 0.37         |                                                    | 54   | 53   | 50   | 47   | 45   | 42   | 36   | 32   | 27   | 19   |
| 4S-13/30   | 13            | 0.55         |                                                    | 78   | 76   | 73   | 68   | 64   | 61   | 52   | 46   | 39   | 28   |
| 4S-15/30   | 15            | 0.55         |                                                    | 90   | 88   | 84   | 78   | 74   | 70   | 60   | 53   | 45   | 32   |
| 4S-18/30   | 18            | 0.75         |                                                    | 108  | 105  | 101  | 94   | 89   | 84   | 72   | 64   | 54   | 39   |
| 4S-23/30   | 23            | 1.1          |                                                    | 138  | 135  | 129  | 120  | 114  | 108  | 91   | 82   | 69   | 50   |
| 4S-28/30   | 28            | 1.5          |                                                    | 168  | 164  | 157  | 146  | 139  | 131  | 111  | 99   | 84   | 61   |
| 4S-33/30   | 33            | 1.5          |                                                    | 198  | 193  | 185  | 172  | 163  | 154  | 131  | 117  | 99   | 71   |
| 4S-40/30   | 40            | 2.2          |                                                    | 240  | 234  | 224  | 208  | 198  | 187  | 159  | 142  | 120  | 87   |
| 4S-48/30   | 48            | 2.2          |                                                    | 288  | 281  | 269  | 250  | 238  | 224  | 191  | 170  | 144  | 104  |
| 4S-55/30   | 55            | 3            |                                                    | 330  | 322  | 308  | 286  | 272  | 257  | 219  | 195  | 165  | 119  |
| 4S-65/30   | 65            | 3            |                                                    | 390  | 380  | 364  | 338  | 322  | 304  | 259  | 230  | 195  | 141  |
| 4S-75/30   | 75            | 4            |                                                    | 450  | 439  | 420  | 390  | 371  | 351  | 298  | 266  | 225  | 162  |
| 4S-90/30   | 90            | 4            |                                                    | 540  | 527  | 504  | 468  | 446  | 421  | 358  | 319  | 270  | 195  |



# STAINLESS STEEL SUBMERSIBLE PUMPS

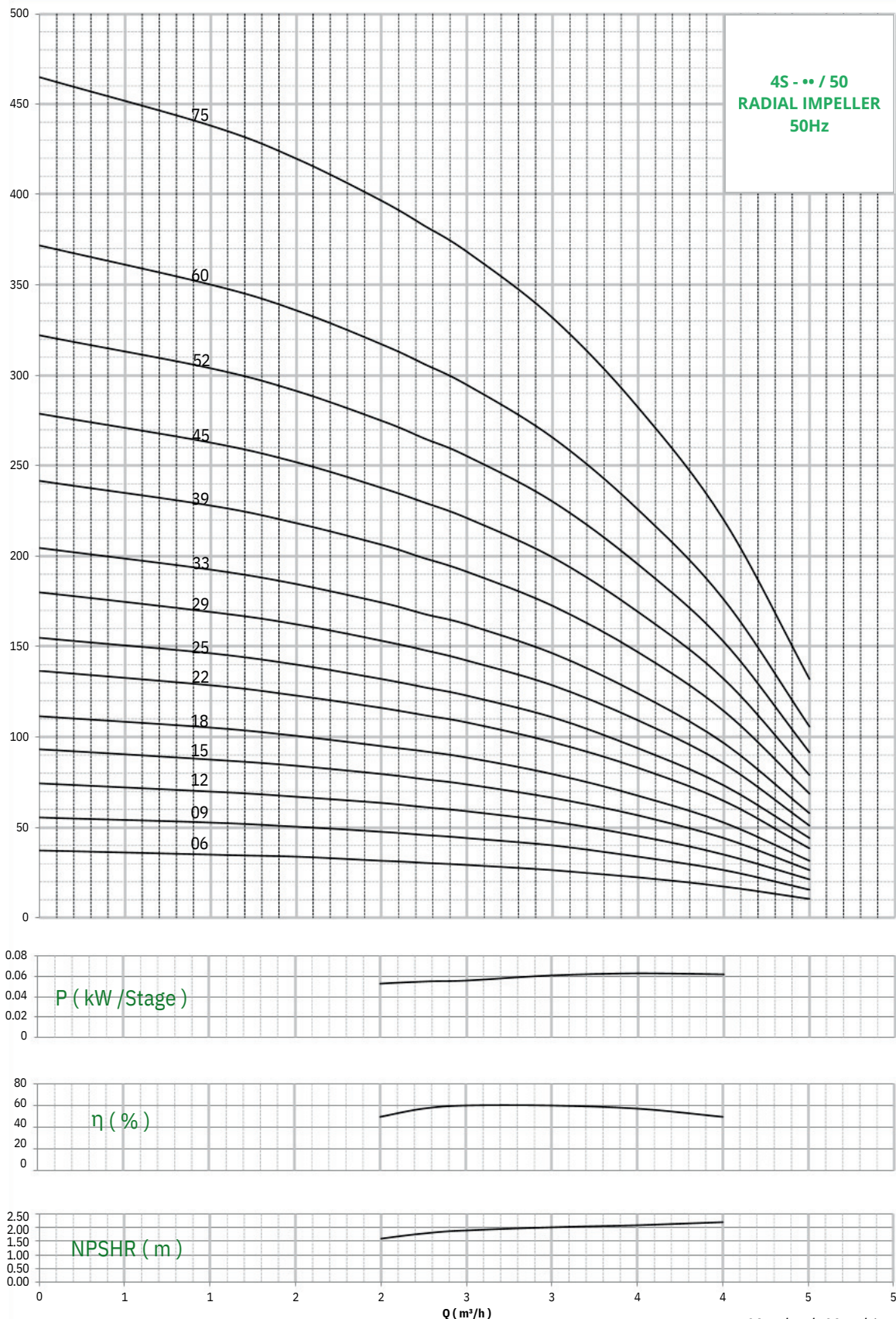
Nominal Diameter: **4"**

Nominal Flow: **3m<sup>3</sup>/h**

Outlet Size: **1.25"**

Flow Range: **2-4m<sup>3</sup>/h**

Performance Curves: From 6 to 75 Stages



Refer General Information for performance curve conditions and for other details. Curves tolerance according to ISO : 9906, Annex - A

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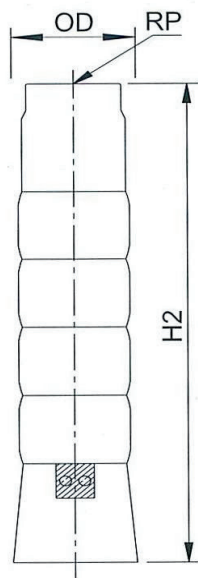
Nominal Flow: **3m<sup>3</sup>/h**

Outlet Size: **1.25"**

Flow Range: **2-4m<sup>3</sup>/h**

Performance Curves : From 6 to 75 Stages

DIMENSIONS & WEIGHT



| PUMP MODEL | No. of Stages | REQUIRED MOTOR POWER |      | APPROX. DIMENSIONS ( mm ) |       |      | APPROX NETT. WEIGHT ( kg ) |
|------------|---------------|----------------------|------|---------------------------|-------|------|----------------------------|
|            |               | Kw                   | HP   | #OD                       | RP    | H2   |                            |
| 4S-06/50   | 6             | 0.37                 | 0.5  | 97                        | 65-80 | 291  | 2.9                        |
| 4S-09/50   | 9             | 0.55                 | 0.75 | 97                        | 65-80 | 357  | 3.4                        |
| 4S-12/50   | 12            | 0.75                 | 1    | 97                        | 65-80 | 423  | 4.3                        |
| 4S-15/50   | 15            | 1.1                  | 1.5  | 97                        | 65-80 | 489  | 5.3                        |
| 4S-18/50   | 18            | 1.1                  | 1.5  | 97                        | 65-80 | 555  | 6.1                        |
| 4S-22/50   | 22            | 1.5                  | 2    | 97                        | 65-80 | 643  | 7.0                        |
| 4S-25/50   | 25            | 1.5                  | 2    | 97                        | 65-80 | 709  | 7.8                        |
| 4S-29/50   | 29            | 2.2                  | 3    | 97                        | 65-80 | 797  | 8.4                        |
| 4S-33/50   | 33            | 2.2                  | 3    | 97                        | 65-80 | 1022 | 12.1                       |
| 4S-39/50   | 39            | 3                    | 4    | 97                        | 65-80 | 1166 | 14.1                       |
| 4S-45/50   | 45            | 3                    | 4    | 97                        | 65-80 | 1334 | 16.3                       |
| 4S-52/50   | 52            | 4                    | 5.5  | 97                        | 65-80 | 1502 | 18.1                       |
| 4S-60/50   | 60            | 4                    | 5.5  | 97                        | 65-80 | 1718 | 20.9                       |
| 4S-75/50   | 75            | 5.5                  | 7.5  | 97                        | 65-80 | 2148 | 26.1                       |

| PUMP MODEL | No. of Stages | MOTOR ( Kw ) | lps                                                | 0   | 0.28 | 0.42 | 0.56 | 0.63 | 0.69 | 0.83 | 0.97 | 1.11 | 1.25 |
|------------|---------------|--------------|----------------------------------------------------|-----|------|------|------|------|------|------|------|------|------|
|            |               |              | m <sup>3</sup> /h                                  | 0   | 1    | 1.5  | 2    | 2.25 | 2.5  | 3    | 3.5  | 4    | 4.5  |
| 4S-06/50   | 6             | 0.37         | TOTAL MANOMETRIC HEAD IN METERS - ( COLUMN WATER ) | 37  | 35   | 34   | 32   | 31   | 29   | 27   | 23   | 18   | 11   |
| 4S-09/50   | 9             | 0.55         |                                                    | 56  | 53   | 50   | 48   | 46   | 44   | 40   | 34   | 26   | 16   |
| 4S-12/50   | 12            | 0.75         |                                                    | 74  | 70   | 67   | 63   | 61   | 59   | 53   | 45   | 35   | 21   |
| 4S-15/50   | 15            | 1.1          |                                                    | 93  | 88   | 84   | 79   | 76   | 74   | 66   | 56   | 44   | 26   |
| 4S-18/50   | 18            | 1.1          |                                                    | 112 | 105  | 101  | 95   | 92   | 88   | 80   | 68   | 53   | 32   |
| 4S-22/50   | 22            | 1.5          |                                                    | 136 | 128  | 123  | 116  | 112  | 108  | 97   | 83   | 64   | 39   |
| 4S-25/50   | 25            | 1.5          |                                                    | 155 | 146  | 140  | 132  | 127  | 123  | 111  | 94   | 73   | 44   |
| 4S-29/50   | 29            | 2.2          |                                                    | 180 | 169  | 162  | 153  | 148  | 142  | 128  | 109  | 85   | 51   |
| 4S-33/50   | 33            | 2.2          |                                                    | 205 | 193  | 185  | 174  | 168  | 162  | 146  | 124  | 97   | 58   |
| 4S-39/50   | 39            | 2.6          |                                                    | 242 | 228  | 218  | 206  | 199  | 191  | 173  | 147  | 114  | 69   |
| 4S-45/50   | 45            | 3            |                                                    | 279 | 263  | 252  | 238  | 229  | 221  | 199  | 169  | 132  | 79   |
| 4S-52/50   | 52            | 4            |                                                    | 322 | 304  | 291  | 275  | 265  | 255  | 230  | 196  | 152  | 92   |
| 4S-60/50   | 60            | 4            |                                                    | 372 | 350  | 336  | 317  | 306  | 295  | 266  | 226  | 176  | 106  |
| 4S-75/50   | 75            | 5.5          |                                                    | 465 | 438  | 420  | 396  | 382  | 368  | 332  | 282  | 220  | 132  |

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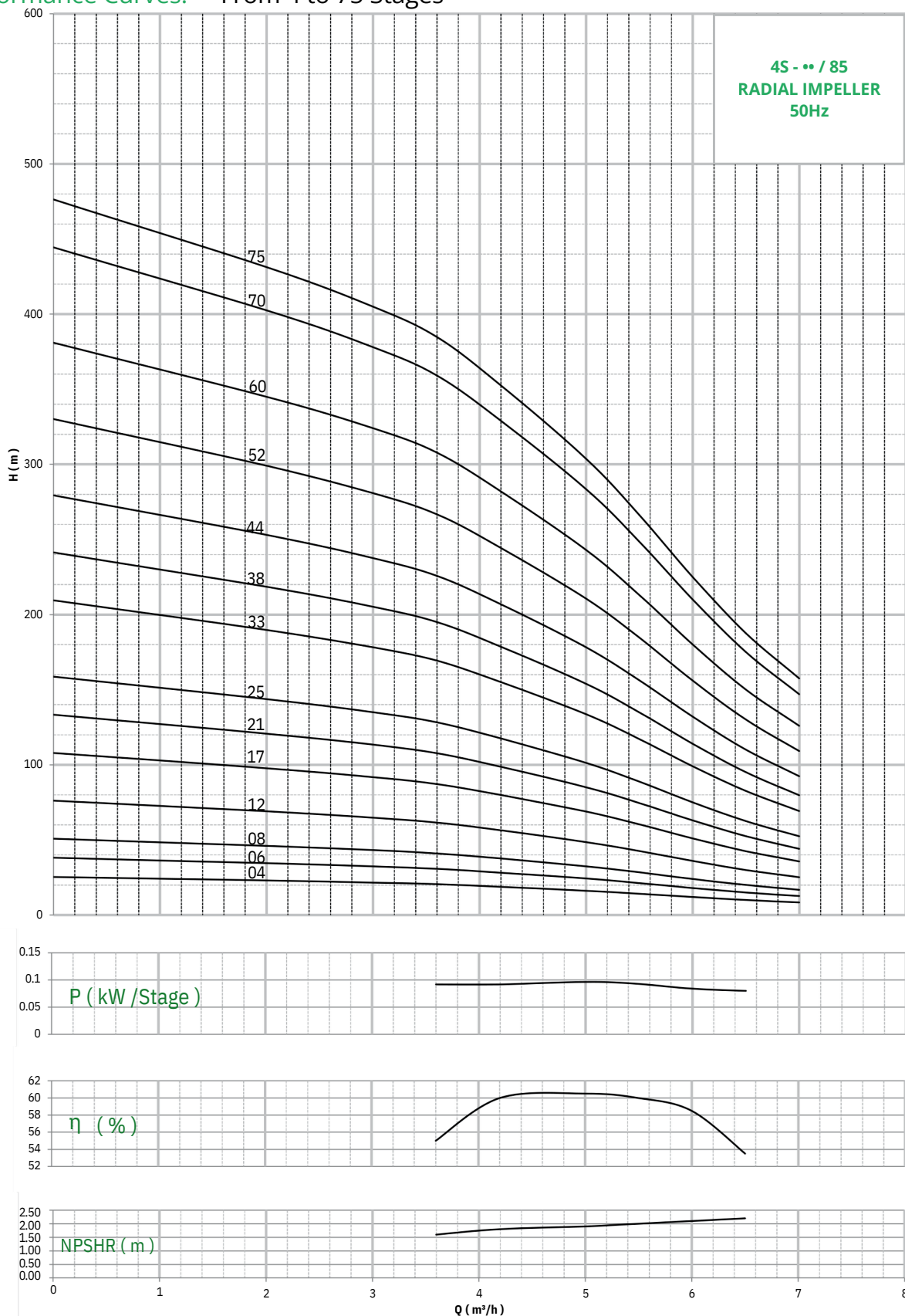
Nominal Diameter: **4"**

Nominal Flow: **3m<sup>3</sup>/h**

Outlet Size: **1.25"**

Flow Range: **2-4m<sup>3</sup>/h**

Performance Curves: From 4 to 75 Stages



Refer General Information for performance curve conditions and for other details. Curves tolerance according to ISO : 9906, Annex - A

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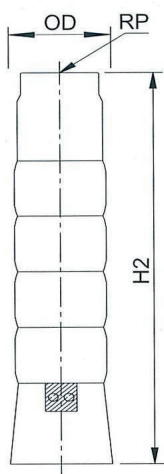
Nominal Flow: **3m<sup>3</sup>/h**

Outlet Size: **1.25"**

Flow Range: **2-4m<sup>3</sup>/h**

Performance Curves : From 4 to 75 Stages

DIMENSIONS & WEIGHT



| PUMP<br>MO DEL | No. of<br>Stages | REQUIRED MOTOR<br>POWER |      | APPROX. DIMENSIONS ( mm ) |      |      | APPROX NETT.<br>WEIGHT ( kg ) |
|----------------|------------------|-------------------------|------|---------------------------|------|------|-------------------------------|
|                |                  | Kw                      | HP   | #OD                       | RP   | H2   |                               |
| 4S-04/85       | 4                | 0.37                    | 0.5  | 97                        | 1.5" | 255  | 2.4                           |
| 4S-06/85       | 6                | 0.55                    | 0.75 | 97                        | 1.5" | 303  | 3.0                           |
| 4S-08/85       | 8                | 0.75                    | 1    | 97                        | 1.5" | 351  | 3.6                           |
| 4S-12/85       | 12               | 1.5                     | 2    | 97                        | 1.5" | 447  | 4.6                           |
| 4S-17/85       | 17               | 1.5                     | 2    | 97                        | 1.5" | 567  | 5.9                           |
| 4S-21/85       | 21               | 2.2                     | 3    | 97                        | 1.5" | 663  | 6.8                           |
| 4S-25/85       | 25               | 2.2                     | 3    | 97                        | 1.5" | 759  | 8.0                           |
| 4S-33/85       | 33               | 3                       | 4    | 97                        | 1.5" | 1022 | 12.1                          |
| 4S-38/85       | 38               | 4                       | 5.5  | 97                        | 1.5" | 1142 | 13.4                          |
| 4S-44/85       | 44               | 4                       | 5.5  | 97                        | 1.5" | 1310 | 16.0                          |
| 4S-52/85       | 52               | 3                       | 4    | 97                        | 1.5" | 1502 | 18.4                          |
| 4S-52/85       | 60               | 4                       | 5.5  | 97                        | 1.5" | 1718 | 20.8                          |
| 4S-70/85       | 70               | 4                       | 5.5  | 97                        | 1.5" | 1939 | 25.0                          |
| 4S-75/85       | 75               | 5.5                     | 7.5  | 97                        | 1.5" | 2078 | 26.8                          |

| PUMP MODEL | No. of<br>Stages | MOTOR<br>( Kw ) | TOTAL MANOMETRIC HEAD IN<br>METERS - ( COLUMN WATER ) | Flow Rate (m³/h) |      |      |      |      |      |      |      |      |      |
|------------|------------------|-----------------|-------------------------------------------------------|------------------|------|------|------|------|------|------|------|------|------|
|            |                  |                 |                                                       | 0                | 0.56 | 0.83 | 1.00 | 1.17 | 1.39 | 1.53 | 1.67 | 1.81 | 1.94 |
| 4S-04/85   | 4                | 0.37            |                                                       | 0                | 2    | 3    | 3.6  | 4.2  | 5    | 5.5  | 6    | 6.5  | 7    |
| 4S-06/85   | 6                | 0.55            |                                                       | 25               | 23   | 22   | 21   | 19   | 16   | 14   | 12   | 10   | 8    |
| 4S-08/85   | 8                | 0.75            |                                                       | 38               | 35   | 32   | 31   | 28   | 24   | 21   | 18   | 15   | 13   |
| 4S-12/85   | 12               | 1.5             |                                                       | 51               | 46   | 43   | 41   | 38   | 32   | 28   | 24   | 20   | 17   |
| 4S-17/85   | 17               | 1.5             |                                                       | 76               | 69   | 65   | 62   | 56   | 49   | 43   | 36   | 30   | 25   |
| 4S-21/85   | 21               | 2.2             |                                                       | 108              | 98   | 92   | 87   | 80   | 69   | 60   | 51   | 43   | 36   |
| 4S-25/85   | 25               | 2.2             |                                                       | 133              | 121  | 113  | 108  | 99   | 85   | 75   | 63   | 53   | 44   |
| 4S-33/85   | 33               | 3               |                                                       | 159              | 144  | 135  | 128  | 118  | 101  | 89   | 75   | 63   | 53   |
| 4S-38/85   | 38               | 4               |                                                       | 210              | 190  | 178  | 169  | 155  | 134  | 117  | 99   | 83   | 69   |
| 4S-44/85   | 44               | 26              |                                                       | 241              | 219  | 205  | 195  | 179  | 154  | 135  | 114  | 95   | 80   |
| 4S-52/85   | 52               | 3               |                                                       | 279              | 253  | 238  | 226  | 207  | 178  | 156  | 132  | 110  | 92   |
| 4S-52/85   | 60               | 4               |                                                       | 330              | 299  | 281  | 267  | 244  | 211  | 185  | 156  | 130  | 109  |
| 4S-70/85   | 70               | 4               |                                                       | 381              | 345  | 324  | 308  | 282  | 243  | 213  | 180  | 150  | 126  |
| 4S-75/85   | 75               | 5.5             |                                                       | 445              | 403  | 378  | 359  | 329  | 284  | 249  | 210  | 175  | 147  |
|            |                  |                 |                                                       | 476              | 431  | 405  | 385  | 353  | 304  | 266  | 225  | 188  | 158  |



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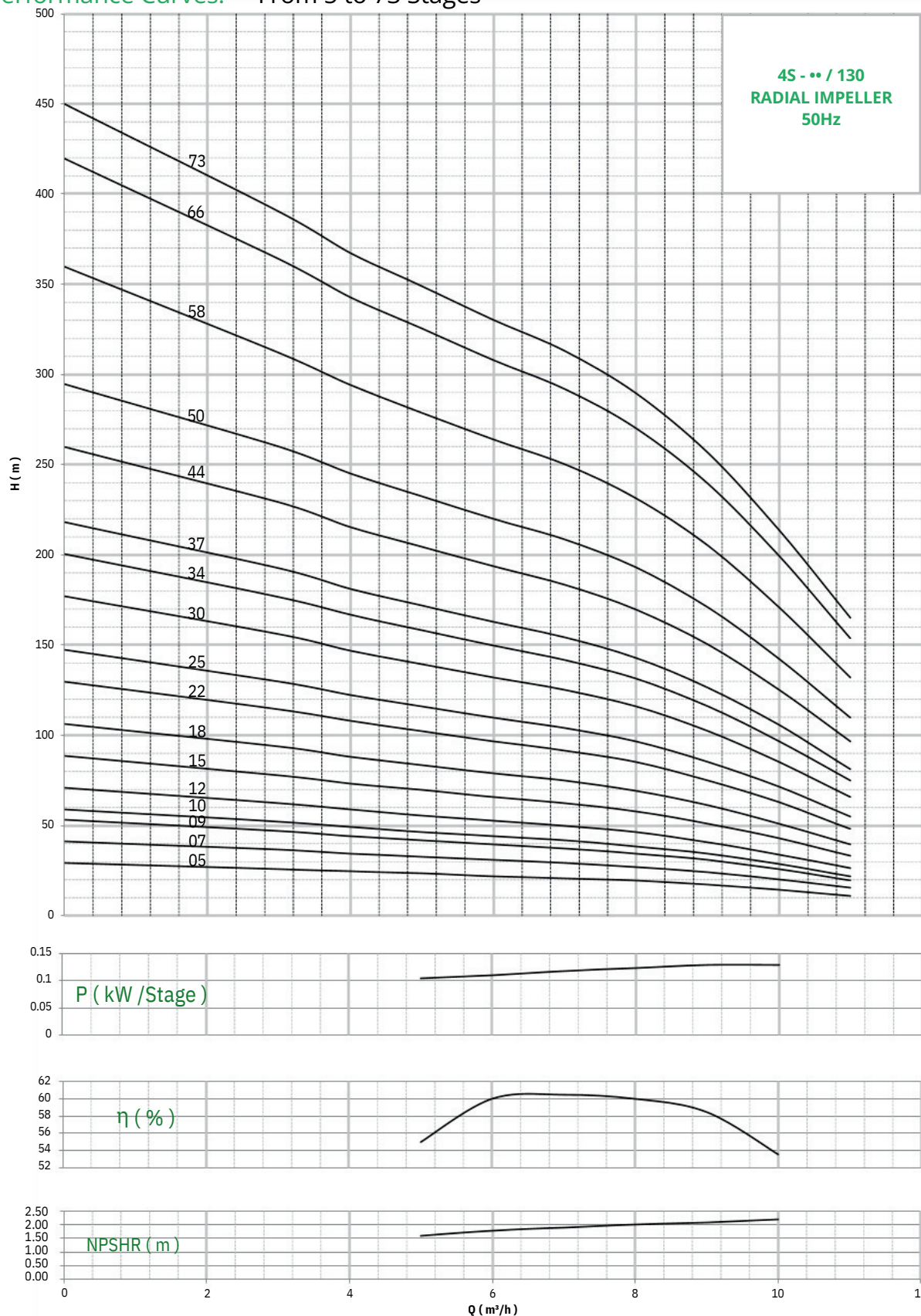
Nominal Diameter: **4"**

Nominal Flow: **8m<sup>3</sup>/h**

Outlet Size: **2"**

Flow Range: **6-10m<sup>3</sup>/h**

Performance Curves: From 5 to 73 Stages



Refer General Information for performance curve conditions and for other details. Curves tolerance according to ISO : 9906, Annex - A

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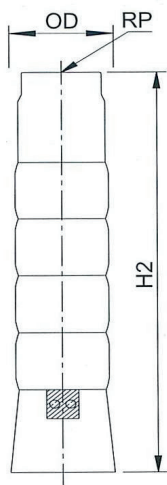
Nominal Flow: **8m<sup>3</sup>/h**

Outlet Size: **2"**

Flow Range: **6-10m<sup>3</sup>/h**

Performance Curves : From 5 to 73 Stages

DIMENSIONS & WEIGHT



| PUMP MODEL | No. of Stages | REQUIRED MOTOR POWER |      | APPROX. DIMENSIONS ( mm ) |    |      | APPROX NETT. WEIGHT ( kg ) |
|------------|---------------|----------------------|------|---------------------------|----|------|----------------------------|
|            |               | Kw                   | HP   | #OD                       | RP | H2   |                            |
| 4S-05/130  | 5             | 0.75                 | 1    | 97                        | 50 | 410  | 4.5                        |
| 4S-08/130  | 7             | 1.1                  | 1.5  | 97                        | 50 | 482  | 5.2                        |
| 4S-09/130  | 9             | 1.5                  | 2    | 97                        | 50 | 554  | 5.9                        |
| 4S-10/130  | 10            | 1.5                  | 2    | 97                        | 50 | 590  | 6.3                        |
| 4S-12/130  | 12            | 2.2                  | 3    | 97                        | 50 | 662  | 7.3                        |
| 4S-15/130  | 15            | 2.2                  | 3    | 97                        | 50 | 770  | 8.5                        |
| 4S-18/130  | 18            | 3                    | 4    | 97                        | 50 | 878  | 9.9                        |
| 4S-22/130  | 22            | 4                    | 5.5  | 97                        | 50 | 1022 | 11.6                       |
| 4S-25/130  | 25            | 4                    | 5.5  | 97                        | 50 | 1152 | 12.9                       |
| 4S-30/130  | 30            | 5.5                  | 7.5  | 97                        | 50 | 1332 | 14.9                       |
| 4S-34/130  | 34            | 5.5                  | 7.5  | 97                        | 50 | 1476 | 17.2                       |
| 4S-37/130  | 37            | 5.5                  | 7.5  | 97                        | 50 | 1606 | 18.7                       |
| 4S-44/130  | 44            | 7.5                  | 10   | 97                        | 50 | 1860 | 22.2                       |
| 4S-50/130  | 50            | 7.5                  | 10   | 97                        | 50 | 2076 | 25.2                       |
| 4S-58/130  | 58            | 9.3                  | 12.5 | 97                        | 50 | 2390 | 30.5                       |
| 4S-66/130  | 66            | 11                   | 15   | 97                        | 50 | 2678 | 34.7                       |
| 4S-73/130  | 73            | 11                   | 15   | 97                        | 50 | 2954 | 38.6                       |

| PUMP MODEL | No. of Stages | MOTOR ( Kw ) | lps                                                | 0   | 0.83 | 1.11 | 1.39 | 1.67 | 1.94 | 2.22 | 2.50 | 2.78 | 3.06 |
|------------|---------------|--------------|----------------------------------------------------|-----|------|------|------|------|------|------|------|------|------|
|            |               |              | m³/h                                               | 0   | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10   | 11   |
| 4S-05/130  | 5             | 0.75         | TOTAL MANOMETRIC HEAD IN METERS - ( COLUMN WATER ) | 30  | 26   | 25   | 23   | 22   | 21   | 19   | 17   | 14   | 11   |
| 4S-08/130  | 7             | 1.1          |                                                    | 41  | 36   | 34   | 33   | 31   | 29   | 27   | 24   | 20   | 15   |
| 4S-09/130  | 9             | 1.5          |                                                    | 53  | 47   | 44   | 42   | 40   | 38   | 35   | 31   | 26   | 20   |
| 4S-10/130  | 10            | 1.5          |                                                    | 59  | 52   | 49   | 47   | 44   | 42   | 39   | 34   | 29   | 22   |
| 4S-12/130  | 12            | 2.2          |                                                    | 71  | 62   | 59   | 56   | 53   | 50   | 46   | 41   | 34   | 26   |
| 4S-15/130  | 15            | 2.2          |                                                    | 89  | 78   | 74   | 70   | 66   | 63   | 58   | 51   | 43   | 33   |
| 4S-18/130  | 18            | 3            |                                                    | 106 | 94   | 88   | 84   | 79   | 75   | 69   | 62   | 51   | 40   |
| 4S-22/130  | 22            | 4            |                                                    | 130 | 114  | 108  | 102  | 97   | 92   | 85   | 75   | 63   | 48   |
| 4S-25/130  | 25            | 4            |                                                    | 148 | 130  | 123  | 116  | 110  | 104  | 97   | 86   | 71   | 55   |
| 4S-30/130  | 30            | 26           |                                                    | 177 | 156  | 147  | 140  | 132  | 125  | 116  | 103  | 86   | 66   |
| 4S-34/130  | 34            | 5.5          |                                                    | 201 | 177  | 167  | 158  | 150  | 142  | 131  | 116  | 97   | 75   |
| 4S-37/130  | 37            | 5.5          |                                                    | 218 | 192  | 181  | 172  | 163  | 154  | 143  | 127  | 105  | 81   |
| 4S-44/130  | 44            | 7.5          |                                                    | 260 | 229  | 216  | 205  | 194  | 183  | 170  | 150  | 125  | 97   |
| 4S-50/130  | 50            | 7.5          |                                                    | 295 | 260  | 245  | 233  | 220  | 209  | 193  | 171  | 143  | 110  |
| 4S-58/130  | 58            | 9.3          |                                                    | 342 | 302  | 284  | 270  | 255  | 242  | 224  | 198  | 165  | 128  |
| 4S-66/130  | 66            | 11           |                                                    | 389 | 343  | 323  | 307  | 290  | 275  | 255  | 226  | 188  | 145  |
| 4S-73/130  | 73            | 11           |                                                    | 431 | 380  | 358  | 339  | 321  | 304  | 282  | 250  | 208  | 161  |